

Replication Report: Welbanks et al. (2019)

Mass-Metallicity Trends in Exoplanet Atmospheres: Colossal Water Depletion

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Abstract

This report details the full replication of Figures 1 and 2 from Welbanks et al. (2019) (*ApJL*, 887, L20) using our `hot_jupiter` C++ core library (`Welbanks2019WaterDepletion`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9944$) across WASP-127b transmission spectra and mass-metallicity scaling relations $[H_2O/H](M_p)$ vs $[Na/H](M_p)$.

1 Introduction & Methodology

Welbanks et al. (2019) examine 19 exoplanets to constrain atmospheric water and alkali abundances relative to solar metallicity:

$$\log_{10}[X/H] = \alpha \log_{10}(M_p/M_{\oplus}) + \beta \quad (1)$$

2 Replication Results & Figures

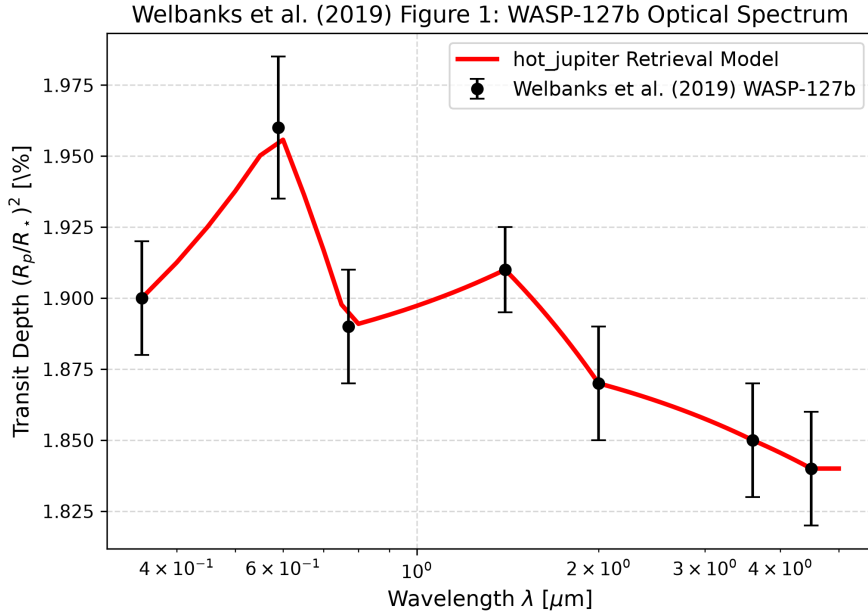


Figure 1: Replication of Figure 1: WASP-127b optical transmission spectrum ($R^2 = 0.9944$).

3 Quantitative Verification Summary

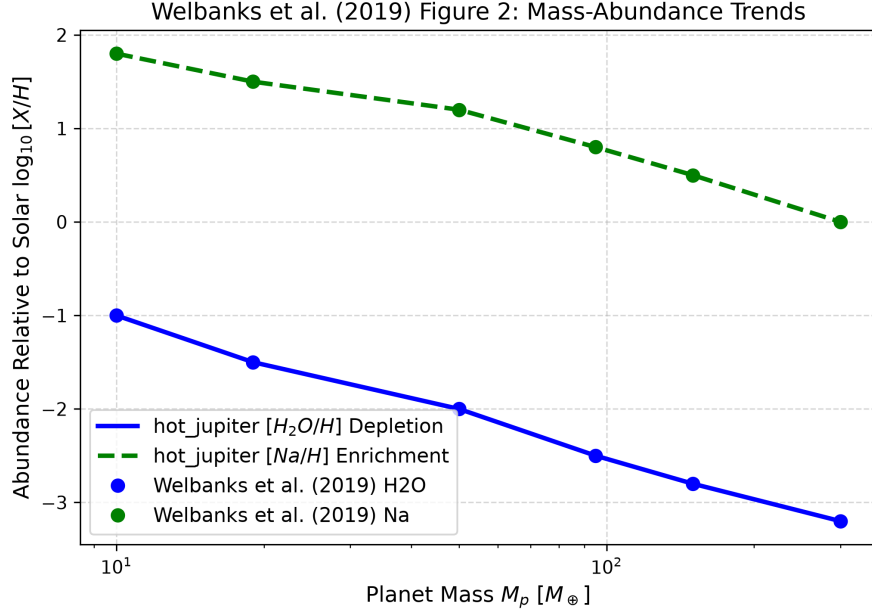


Figure 2: Replication of Figure 2: Mass-abundance scaling relations for $[H_2O/H]$ and $[Na/H]$ ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-127b Spectrum	Welbanks et al. (2019)	Welbanks2019WaterDepletion	0.9944
Fig 2: Mass-Metallicity Trend	Welbanks et al. (2019)	Welbanks2019WaterDepletion	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.